T.Y.B.Sc. (Computer Science) Practical Answers

Savitribai Phule Pune University | SEM V

Lab Course CS-306-MJ-P: Core Java and Web Technology-I

Q1) Java Program — Account, SavingAccount and AccountDetail (15 Marks)

Save as **AccountDetailProgram.java**. This program defines Account with default and parameterized constructors, then extends it through SavingAccount and AccountDetail.

```
class Account {
    String custname;
    int accno;

    // Default constructor
    Account() {
        custname = "Unknown";
        accno = 0;
    }

    // Parameterized constructor
    Account(String name, int number) {
        custname = name;
        accno = number;
    }
}

class SavingAccount extends Account {
    double savingbal;
    double minbal;

    SavingAccount() {
        super();
        savingbal = 0;
        minbal = 1000;
    }

    SavingAccount(String name, int number,
                  double balance, double minimum) {
        super(name, number);
        savingbal = balance;
        minbal = minimum;
    }
}

class AccountDetail extends SavingAccount {
    double depositamt;
    double withdrawalamt;

    AccountDetail(String name, int number,
                  double balance, double minimum,
                  double deposit, double withdrawal) {
        super(name, number, balance, minimum);
        depositamt = deposit;
        withdrawalamt = withdrawal;
    }

    void display() {
        System.out.println("Customer Name: " + custname);
        System.out.println("Account Number: " + accno);
        System.out.println("Saving Balance: " + savingbal);
        System.out.println("Minimum Balance: " + minbal);
        System.out.println("Deposit Amount: " + depositamt);
        System.out.println("Withdrawal Amount: " + withdrawalamt);
        System.out.println("Final Balance: "
                + (savingbal + depositamt - withdrawalamt));
    }
}

public class AccountDetailProgram {
    public static void main(String[] args) {
        AccountDetail a = new AccountDetail(
                "Yash", 101, 5000, 1000, 2000, 500);
        a.display();
```

```
    }
}
```

Expected output

```
Customer Name: Yash
Account Number: 101
Saving Balance: 5000.0
Minimum Balance: 1000.0
Deposit Amount: 2000.0
Withdrawal Amount: 500.0
Final Balance: 6500.0
```

Explanation: Account stores the customer name and account number. SavingAccount inherits Account and stores the saving and minimum balances. AccountDetail inherits SavingAccount and stores deposit and withdrawal amounts. The display() method prints the details. Final balance = saving balance + deposit – withdrawal.

Q2) HTML Program — Button and Link Hover Effect (15 Marks)

Save the following code as **index.html** and open it in a web browser.

```
<!DOCTYPE html>
<html>
<head>
    <title>Hover Effect</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            text-align: center;
            margin-top: 100px;
        }

        button {
            background-color: blue;
            color: white;
            padding: 12px 25px;
            border: none;
            border-radius: 5px;
            cursor: pointer;
            font-size: 18px;
        }

        button:hover {
            background-color: green;
            color: yellow;
        }

        a {
            display: inline-block;
            margin-top: 30px;
            color: blue;
            text-decoration: none;
            font-size: 20px;
        }

        a:hover {
            color: red;
            text-decoration: underline;
        }
    </style>
</head>
<body>
    <h2>CSS Hover Effect</h2>
    <button>Click Me</button>
    <br>
    <a href="https://www.google.com">Visit Google</a>
</body>
</html>
```

Expected behavior

Before hover: the button is blue with white text, and the link is blue. When the pointer moves over the button, its background turns green and its text turns yellow. When the pointer moves over the link, the text turns red and becomes underlined. When the pointer moves away, the original styles return.

Explanation

The :hover CSS pseudo-class applies styles when the mouse pointer is over an element. Example: `button:hover { background-color: green; }`. The <button> tag creates a button, and the <a> tag creates a hyperlink.

Q3) Viva Questions and Answers (5 Marks)

1. What is Java?
Java is a high-level, object-oriented programming language.

2. What is a class?
A class is a blueprint for creating objects.

3. What is an object?
An object is an instance of a class.

4. What is a constructor?
A constructor is a special member used to initialize an object.

5. What is a default constructor?
A no-argument constructor is called a no-argument constructor. Java provides a default constructor only when no constructor is declared.

6. What is a parameterized constructor?
A constructor that accepts arguments is called a parameterized constructor.

7. What is inheritance?
Inheritance allows one class to acquire properties and methods of another class.

8. Which keyword is used for inheritance in Java?
The extends keyword.

9. What is the super keyword?
super is used to access parent-class constructors and members.

10. What is a subclass?
A subclass is a class that inherits from another class.

11. What is HTML?
HTML stands for HyperText Markup Language. It structures web pages.

12. What is CSS?
CSS stands for Cascading Style Sheets. It styles web pages.

13. What is a pseudo-class?
A pseudo-class defines a special state of an HTML element.

14. What is :hover in CSS?
:hover applies styles when the mouse pointer is over an element.

15. What is a hyperlink?
A hyperlink connects to another webpage or resource.

16. Which HTML tag creates a hyperlink?
The <a> tag.

17. Which HTML tag creates a button?
The <button> tag.

18. What is an HTML attribute?
An attribute provides additional information about an HTML element.

19. What is the <style> tag?
The <style> tag contains internal CSS.

20. What is the difference between HTML and CSS?
HTML creates webpage structure; CSS controls its appearance.

Quick revision: Q1 — classes, constructors, inheritance, super(), display method. Q2 — HTML, CSS, button, hyperlink, :hover. Q3 — Java basics, inheritance, HTML and CSS.